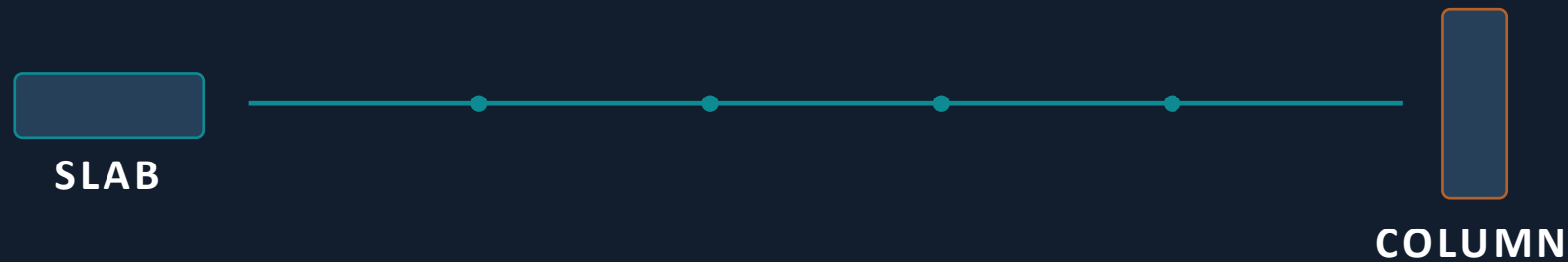


Application of High-Resolution Distributed Optical Fibre in Damage Assessment of Ageing RC Structures

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Ageing Reinforced-Concrete infrastructures

Three pressures acting at once on a stock that was never designed for them.



Increasing traffic volumes and heavier axle loads



Progressive environmental deterioration like corrosion



Material ageing and cracking over time



RESEARCH GAP AND SIGNIFICANCE

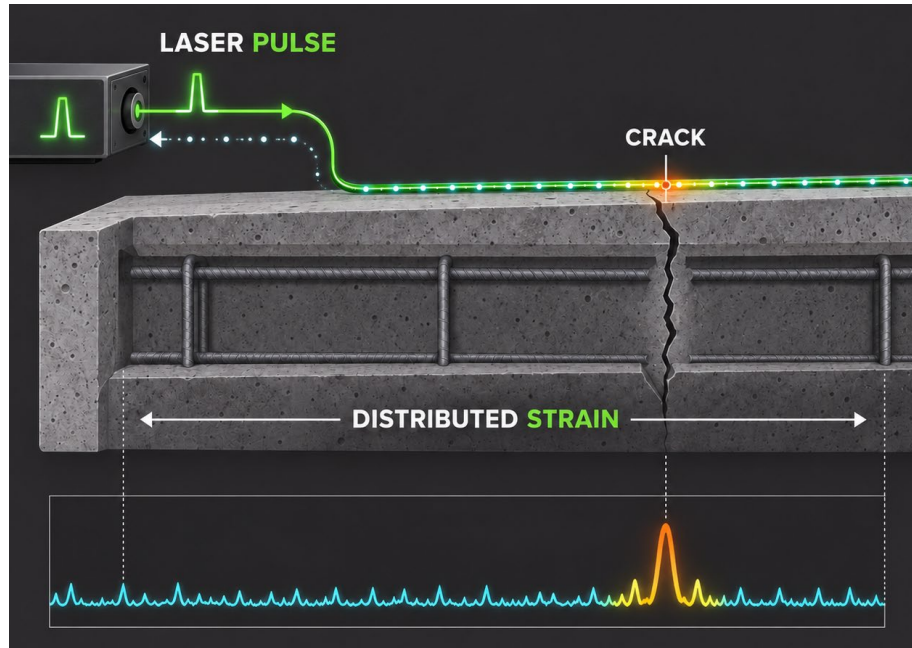
- Visual inspection and point sensors sample a handful of locations, damage between them goes unseen.
- Global indicators move little while local stress rises sharply.
- Few experimental datasets capture corroded RC members at high spatial resolution.

WHAT THIS DEMANDS

A way to measure strain everywhere along a member, not at a few chosen points.

One Fibre, Thousands of Measurement Points

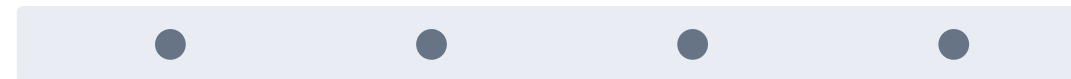
High-resolution distributed acoustic sensing (HR-DAS) turns a single glass fibre into a continuous line of strain sensors.



Strain is resolved all along the fibre, so a crack appears as a local peak.

Strain Resolution	0.001 $\mu\epsilon$
Spatial resolution	20 mm
Sensing Range	2 km
Number of Sensing Nodes	10,000
Max Ping Rate	500 kHz

sensors at a few fixed points



a distributed fibre, measuring everywhere along its length

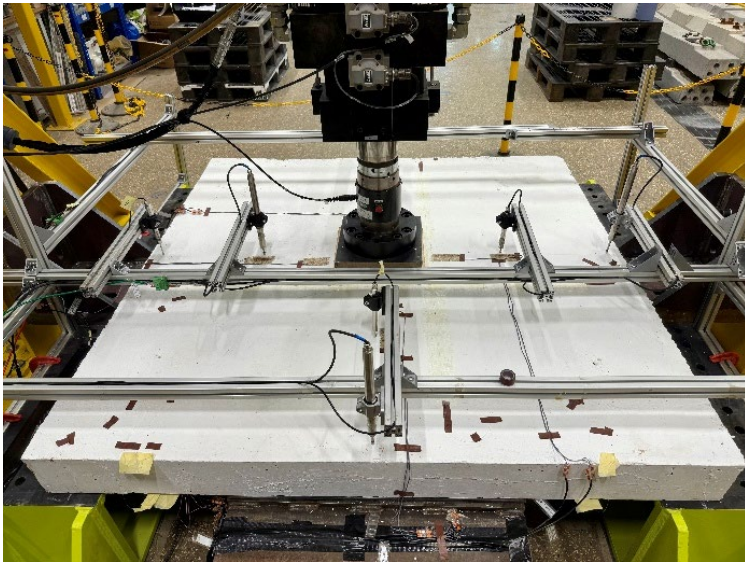


THE AIM OF THIS WORK

Here we use this technique to track corrosion damage in RC elements as it develops.

Two Studies, Two Different RC Elements

TWO-WAY SLAB



- 1700 × 1700 × 150 mm two-way slab
- Concrete 46.5 MPa (cube strength)
- Reinforcement yield strength 524 Mpa

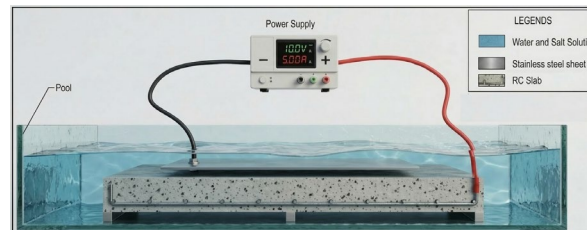
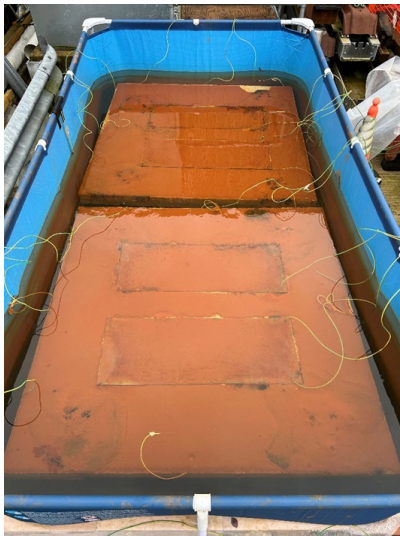
RC COLUMN



- 300 × 300 mm cross-section
- 1500 mm height

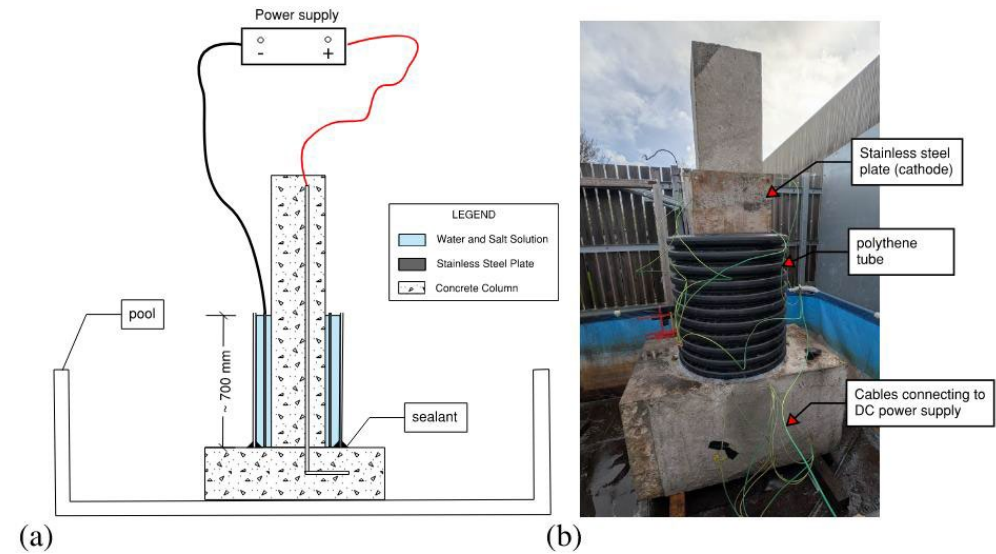
SLAB CORROSION

Impressed-current corrosion, accelerating reinforcement corrosion within a practical laboratory timeframe.



COLUMN CORROSION

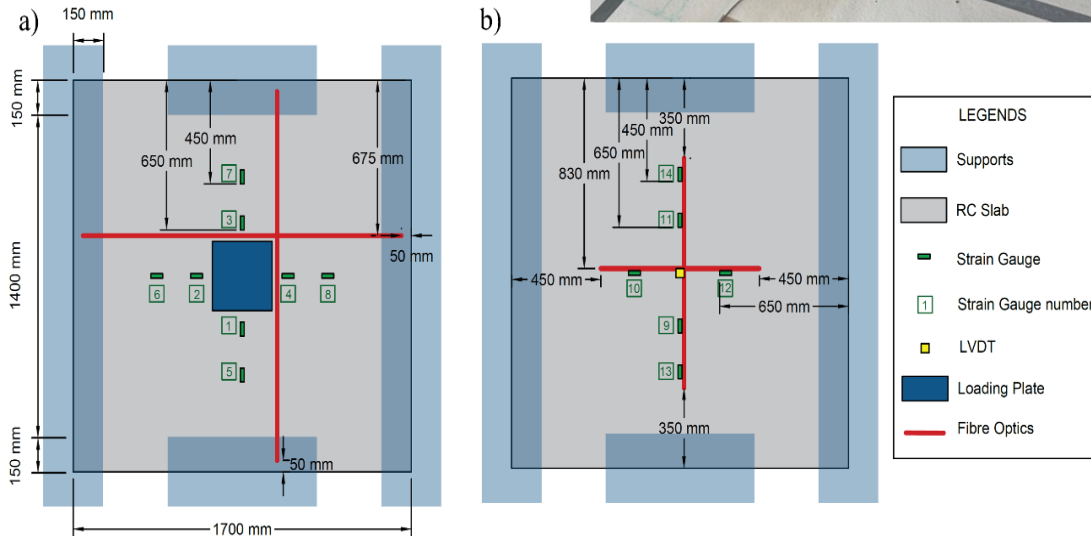
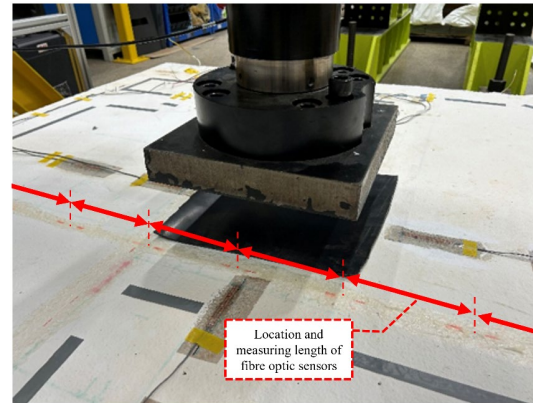
Constant impressed current applied to the lower part of the column, the critical plastic-hinge region.



How the Distributed Fibre Was Placed

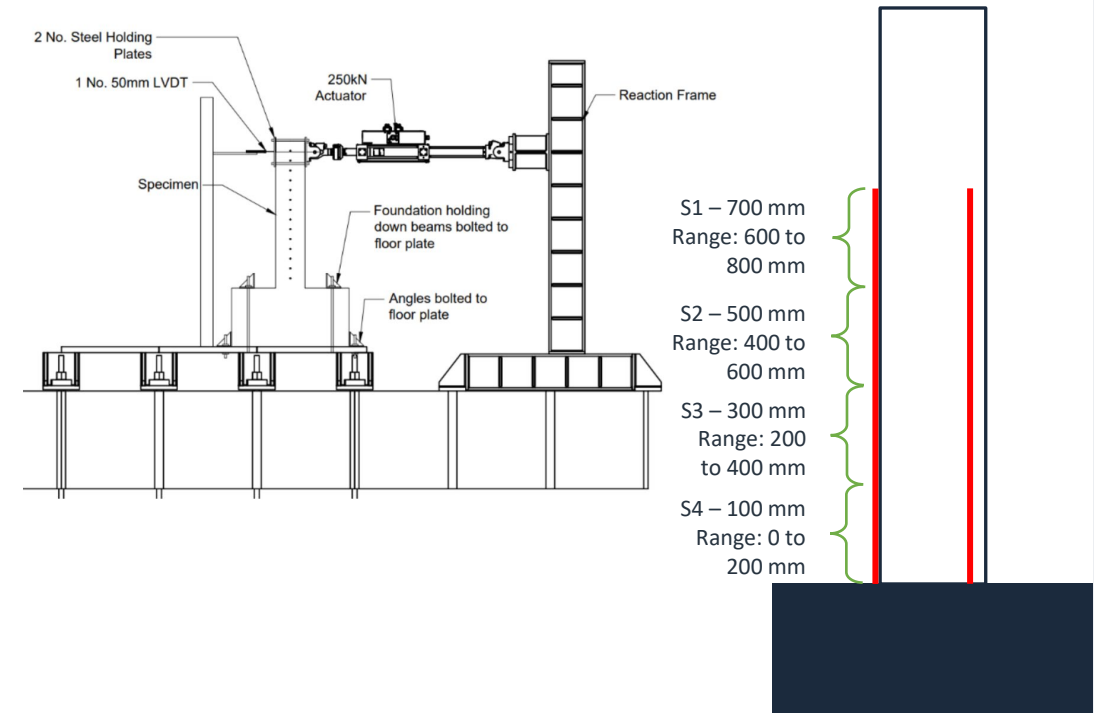
SLAB: SURFACE SENSING

- Fibre bonded top & bottom surfaces, both directions.



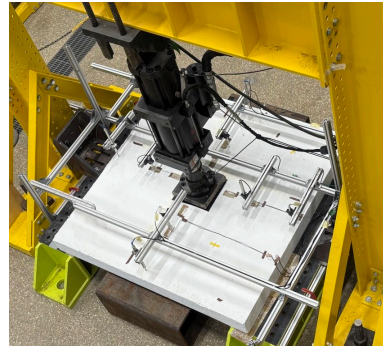
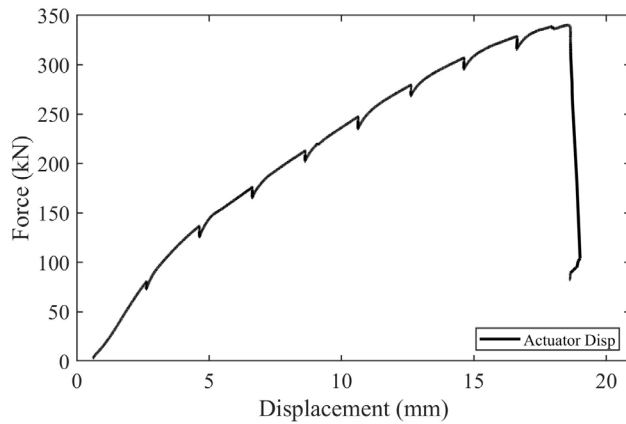
COLUMN: SURFACE + REBAR SENSING

- External fibre: bonded to the concrete surface.
- Internal fibre: bonded to a longitudinal bar.



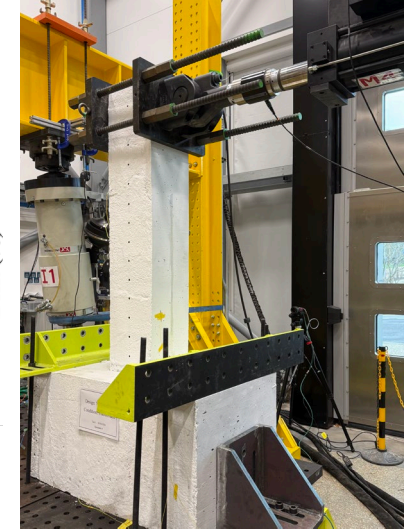
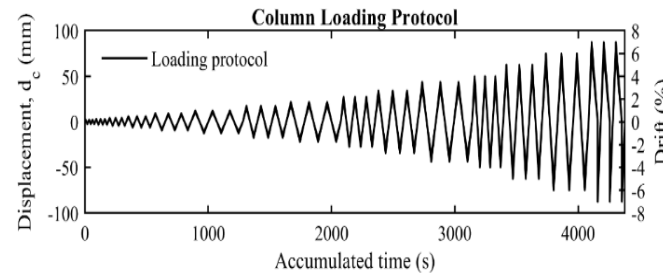
Two Elements, Two Loading Regimes

MONOTONIC CENTRAL LOADING



- 500 kN hydraulic actuator
- Displacement control at 1 mm/min
- Paused every 2 mm to mark cracks and run a hammer test

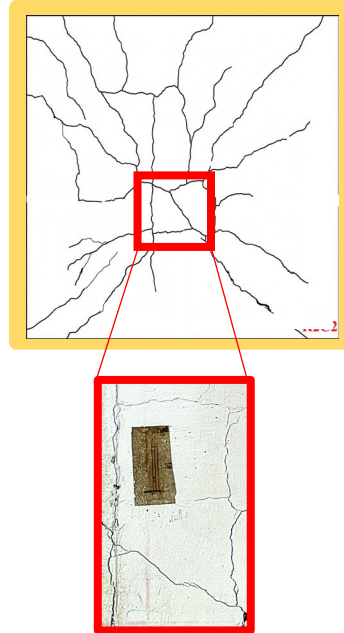
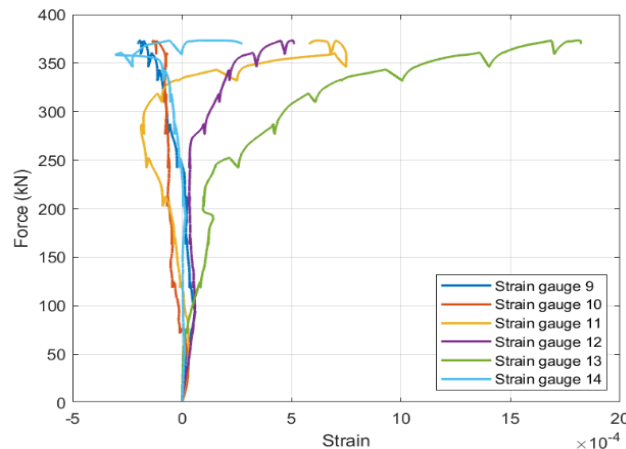
REVERSED-CYCLIC LOADING



- 250 kN actuator, lateral displacement under displacement control
- Lateral displacements from 2.1 to 73.5 mm, three repeated cycles at each deformation level (ACI 374.1-05)
- Testing continued to 7% drift.

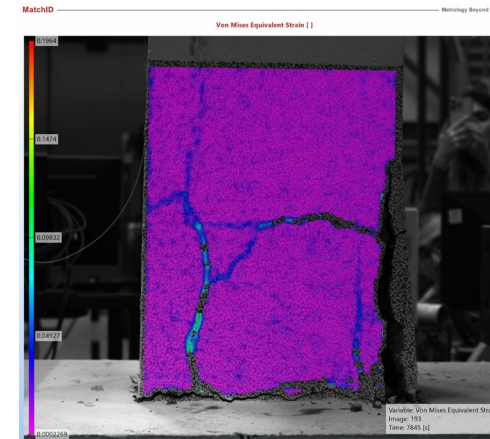
Why Strain Gauges and DIC Are Not Enough

STRAIN GAUGES

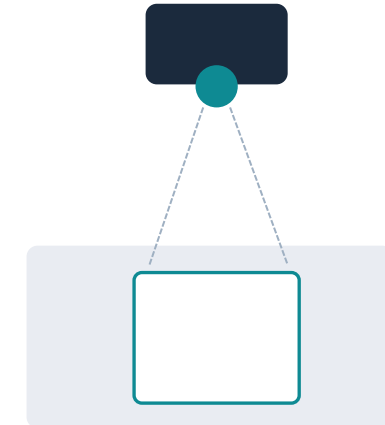


- Cracks crossing a gauge can cause abrupt readings, debonding or sensor damage.
- Concrete between cracks may show strain relief as cracking progresses.
- Measurement positions are fixed; cracks may develop between gauges.

DIGITAL IMAGE CORRELATION



fixed high-resolution camera

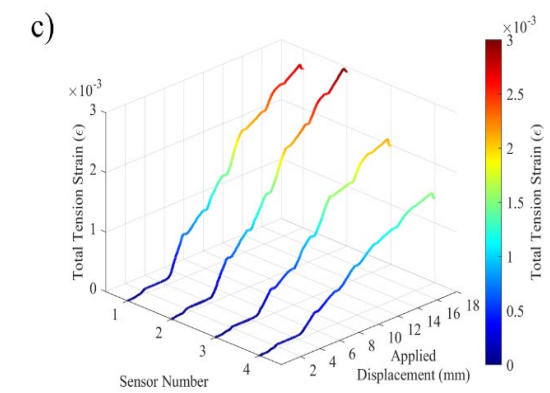
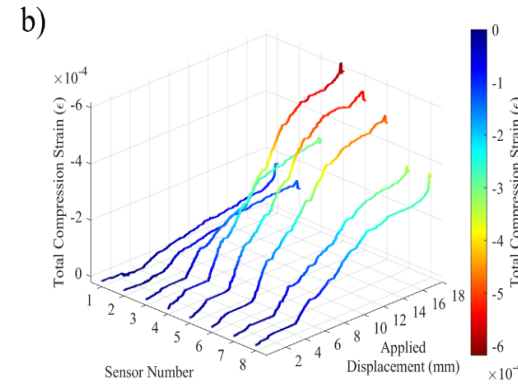
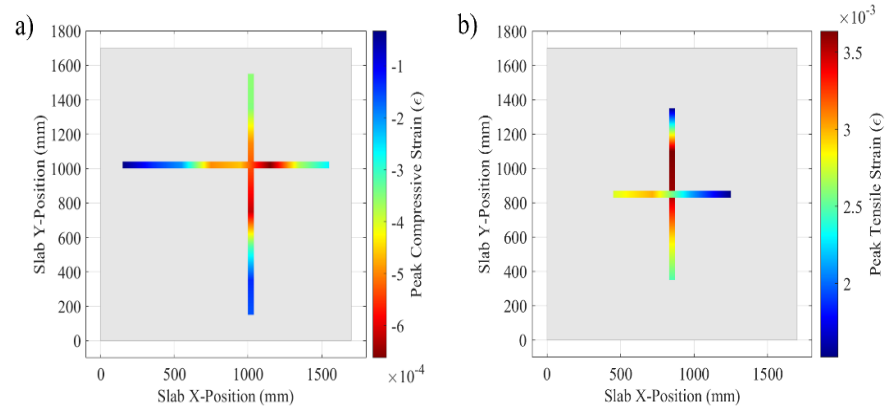


small pre-selected patch

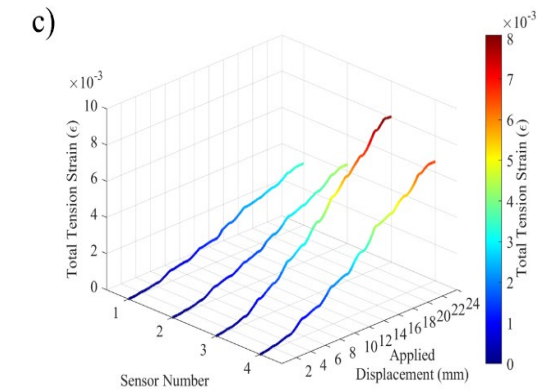
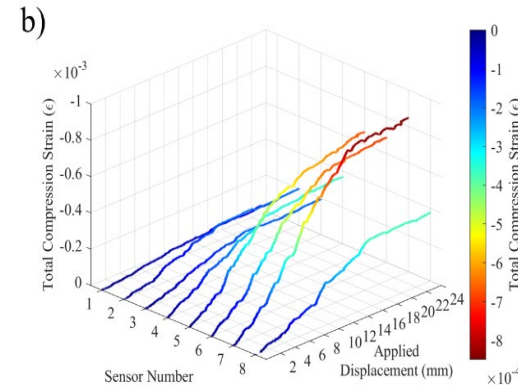
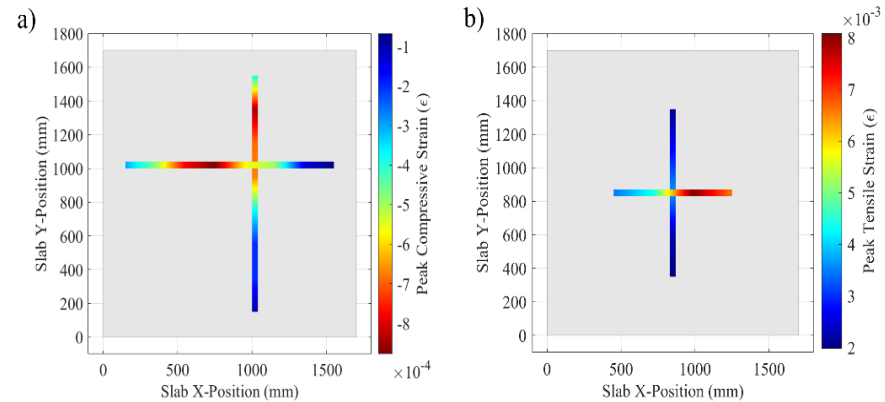
- Full-field measurements within a preselected, visible surface area.
- Requires suitable surface texture, clear line of sight, suitable lighting and a stable camera position.
- Maintaining these conditions can be challenging for long-term field monitoring.

Distributed Strain Along the Slab Sensing Paths

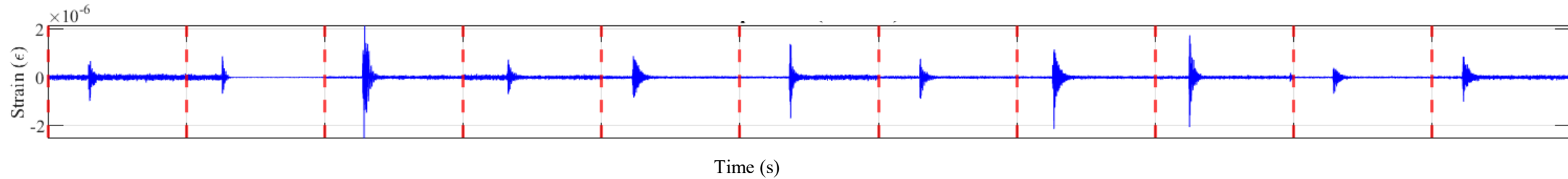
Uncorroded



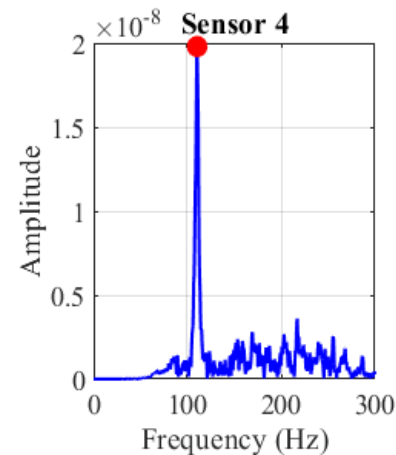
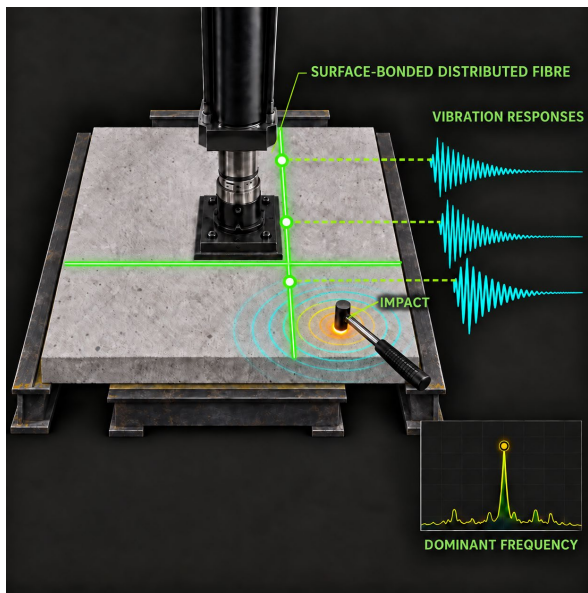
Corroded



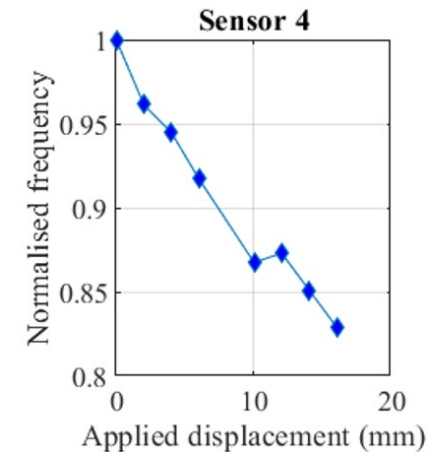
Peak compressive & tensile strain maps (left) and 3D time-history evolution, top & bottom surface, X-axis (right)



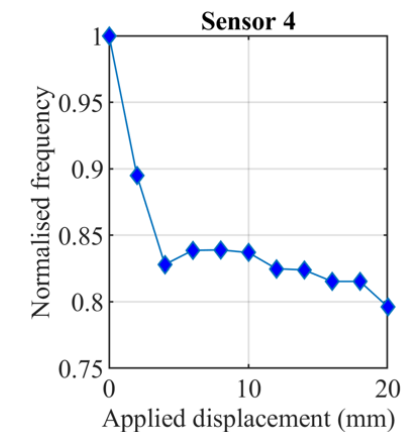
- Time-history strain response recorded by Sensor 4 during hammer excitations after each step of loading



- Fourier amplitude spectrum of the hammer-test response



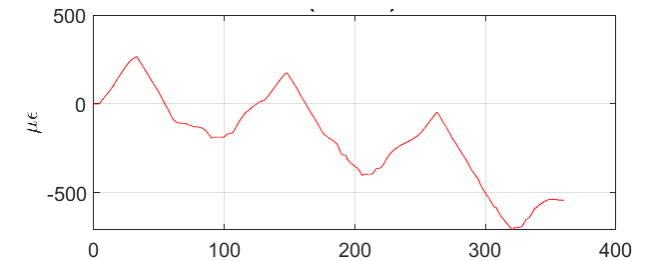
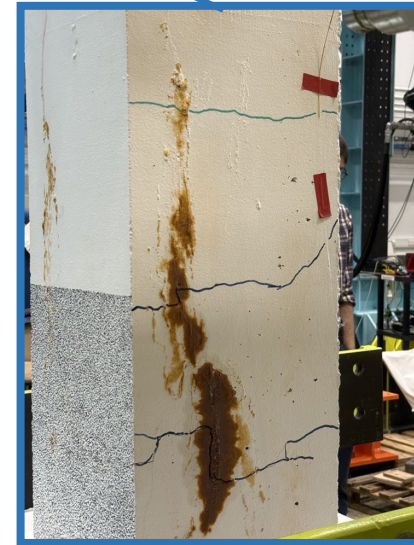
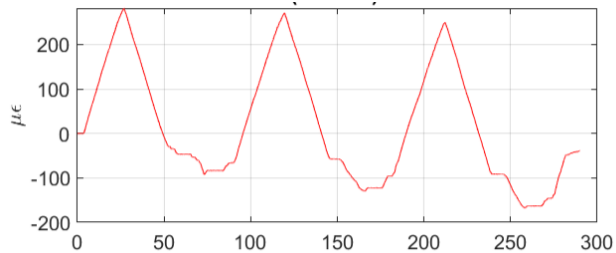
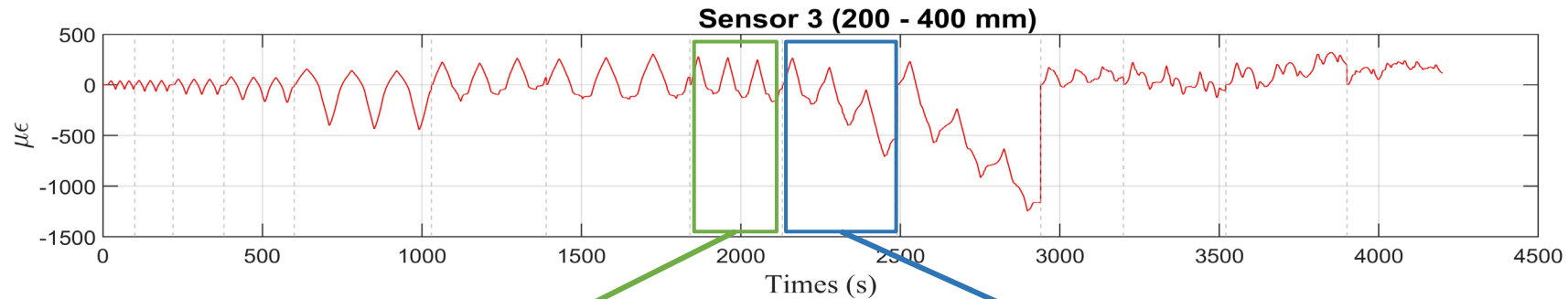
- Evolution of normalised dominant frequency with applied displacement for the **uncorroded** slab



- Evolution of normalised dominant frequency with applied displacement for the **corroded** slab

Corrosion → earlier frequency reduction → earlier stiffness degradation

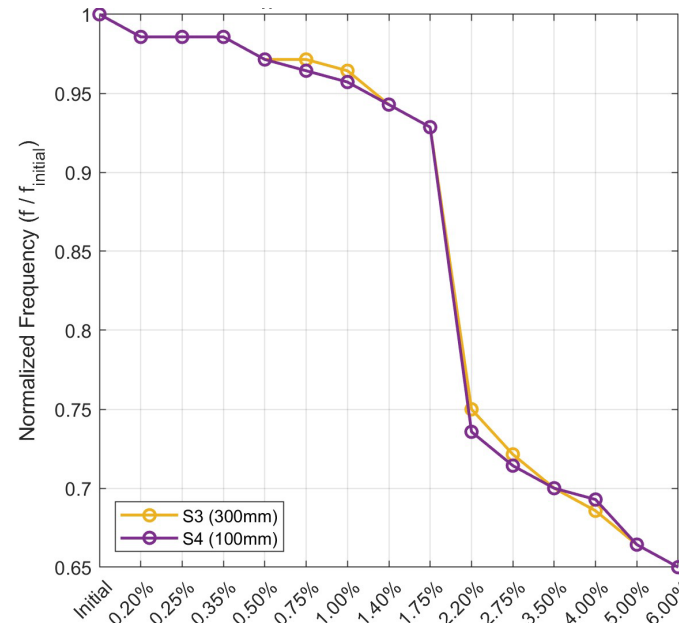
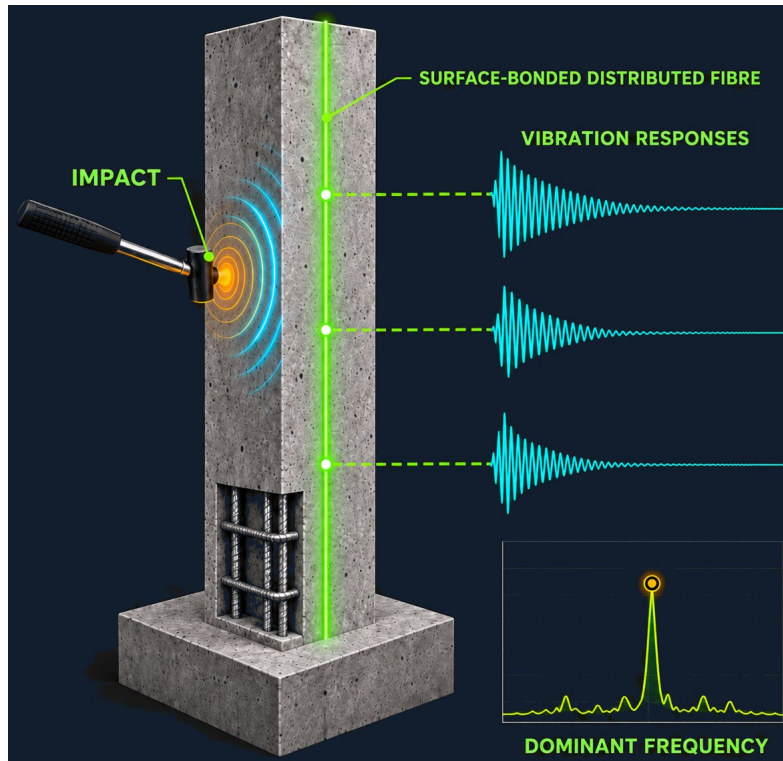
Column Strain Response During Cyclic Loading.



- Fibre measurements track changes in the strain response throughout cyclic loading.
- Cycles become less regular and strain levels shift as cracking develops.

Fibre-Based Vibration Monitoring Identifies a Major Damage Transition

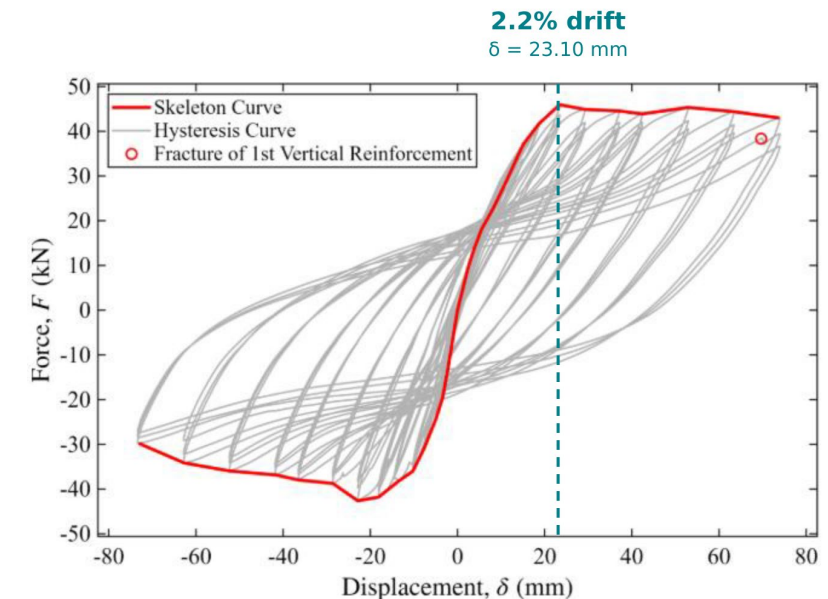
The same installed fibre, used dynamically: one hammer impact after each loading step.



Normalised dominant frequency at two sensing positions along the column

2.2% drift

sharp frequency reduction, coinciding with peak lateral strength



WHAT THE COLUMN SHOWS

One sharp transition, captured consistently at critical sensing position.

Towards Long-Term Bridge Monitoring

Follow structural changes

Track strain and vibration response against a baseline over time.

Target inspection and repair

Identify critical regions that need closer assessment.

Support service-life decisions

Use monitoring evidence to inform maintenance and repair decisions.



Long-term evidence to support maintenance and safe service-life extension

Field Monitoring: Challenges and Next Steps

CHALLENGES

Durable installation

Maintain bonding and cable protection through long-term exposure.

Environmental effects

Distinguish damage-related changes from environmental (like temperature effects) and load effects.

Monitoring at scale

Balance coverage, resolution and sampling rate with cost and data management.

Reliable interpretation

Distinguish structural changes from measurement problems.

NEXT STEPS

Test installation durability

Evaluate bonds and protection systems under realistic exposure conditions.

Account for field conditions

Use temperature compensation and account for changes in loading.

Design practical monitoring systems

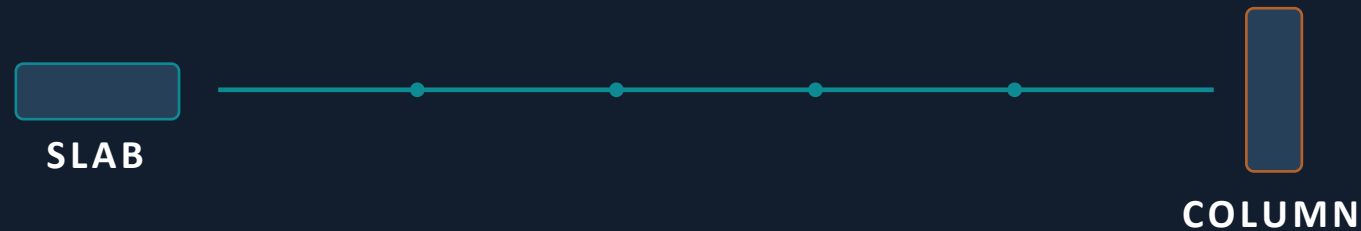
Select critical fibre routes and plan power, data handling and maintenance.

Validate damage indicators

Check measured indicators against inspections and structural assessment.

Thank You

Questions?



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